MIL-S-19500/162B
AMENDMENT 4
29 July 1974
SUPERSEDING
AMENDMENT 3
4 November 1966

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER

TYPES 1N1614, 1N1615, 1N1616, 1N4458, 1N4459,
1N1614R, 1N1615R, 1N1616R, 1N4458R, 1N4459R

This amendment forms a part of Military Specification MIL-S-19500/162B, dated 27 July 1964, and is approved for use by all Departments and Agencies of the Department of Defense.

Page 1

1.3, i_f(surge) column: Delete "150" and substitute "100".

1.3, BAROMETRIC PRESSURE, REDUCED: Delete "8 mm Hg at $V_{RM} \leq 600$ Vdc, 15 mm Hg at $V_{RM} \geq 600$ Vdc" and substitute "8 mmHg - 1N1614 and 1N1615; 16 mmHg - 1N1616; 30 mmHg - 1N4458; 54 mmHg - 1N4459".

Page 3

4.1, delete and substitute:

"4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein, except that the lot accumulation period requirements shall be 6 months in lieu of 6 weeks."

*4.3, delete and substitute:

"4.3 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in groups A, B, and C. Group A inspection shall be performed on a sublot basis. Group B inspection shall be performed on a lot basis per MIL-S-19500. Subgroups 7 and 8 of group B and subgroups 1, 2, and 3 of group C inspection shall be performed on a sublot basis or on the highest voltage type present in the lot. In the event subsequent lots contain higher voltage devices than have previously been accepted under group C provisions, these higher voltage devices shall be subjected to group C inspection. A failure of these higher voltage devices does not invalidate previously established group C status on lower voltage devices."

*Add the following new paragraph:

"4.3.4 Group B and group C life-test samples. Samples that have been subjected to group B 340-hours life-test, may be continued on test to 1,000 hours to satisfy group C life-test requirements. These samples shall be predesignated and shall remain subjected to the group C 1,000-hour acceptance evaluation after they have passed the group B 340-hour acceptance criteria. The cumulative total of failures found during 340-hour test and during the subsequent interval up to 1,000 hours shall be computed for 1,000-hour acceptance criteria (see 4.3.3)."

4.4.1, delete and substitute:

"4.4.1 Time limit for end points. End-point tests for qualification and quality-conformance inspection shall be completed within 96 hours after completion of the last test in the subgroup."

Page 4

TABLE I, Subgroup 2: Delete "Forward voltage" test and substitute:

"

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Forward voltage	4011	i_f = 15 a(pk); $t_p \leq$ 8.3 msec, duty cycle $\leq$ 2%, pulse		v_f	—	1.5	v(pk)

"

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TABLE II, Subgroup 2: Delete "End points" and "Forward voltage" and substitute:

"

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
End points: (See 4.4.1)							
Forward voltage	4011	i_f = 15 a(pk); $t_p \leq$ 8.3 msec, duty cycle $\leq$ 2%, pulse		v_f	—	1.5	v(pk)

"

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TABLE II, Subgroup 5, Surge current, Details column: Delete "i_f(surge) = 150 a" and substitute "i_f(surge) = 100 a".

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*TABLE II, Subgroups 7, 8, and 9, delete and substitute:

"

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 7			10				
High-temperature life (nonoperating	1032	T_A = 200°C (see 4.3.4)		---	—	—	—
End points: (See 4.4.1)							
Forward voltage:	4011	i_f = 15 a(pk); $t_p \leq$ 8.3 msec, duty cycle $\leq$ 2%, pulse		v_f	—	1.5	v(pk)
Reverse current:	4016			I_R			
1N1614		V_R = 200 Vdc			—	100	µAdc
1N1615		V_R = 400 Vdc			—	100	µAdc
1N1616		V_R = 600 Vdc			—	100	µAdc
1N4458		V_R = 800 Vdc			—	100	µAdc
1N4459		V_R = 1,000 Vdc			—	100	µAdc
Subgroup 8			10				
Steady state operation: (see 4.4.2)	1027	T_C = 150°C; I_o = 5 Adc; f = 60 Hz					
1N1614		V_{RM}(wkg) = 200 v(pk)			—	—	—
1N1615		V_{RM}(wkg) = 400 v(pk)			—	—	—
1N1616		V_{RM}(wkg) = 600 v(pk)			—	—	—
1N4458		V_{RM}(wkg) = 800 v(pk)			—	—	—
1N4459		V_{RM}(wkg) = 1,000 v(pk)			—	—	—
End points: (Same as subgroup 7)							

"

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Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 9			10				
Low-temperature operation:		T_A = -65°C					
Forward voltage:	4011	i_f = 15 a(pk); $t_p \leq$ 8.3 msec, duty cycle ≤ 2%, pulse		v_f	—	1.7	v(pk)
Reverse current:	4016			I_R			
1N1614		V_R = 200 Vdc			—	100	µAdc
1N1615		V_R = 400 Vdc			—	100	µAdc
1N1616		V_R = 600 Vdc			—	100	µAdc
1N4458		V_R = 800 Vdc			—	100	µAdc
1N4459		V_R = 1,000 Vdc			—	100	µAdc

*TABLE III, delete and substitute:

TABLE III. Group C inspection.

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1 1/			10				
Barometric pressure, reduced (altitude operation)	1001	8 mmHg - 1N1614 and 1N1615; 16 mmHg - 1N1616; 30 mmHg - 1N4458; 54 mmHg - 1N4459; t = 60 sec		—	—	—	—
Measurement during test:							
Reverse current:	4016			I_R			
1N1614		V_R = 200 Vdc			—	50	µAdc
1N1615		V_R = 400 Vdc			—	50	µAdc
1N1616		V_R = 600 Vdc			—	50	µAdc
1N4458		V_R = 800 Vdc			—	50	µAdc
1N4459		V_R = 1,000 Vdc			—	50	µAdc
Subgroup 2			λ=10				
High-temperature life (nonoperating)	1031	T_A = 200°C (see 4.3.4)		—	—	—	—
End points: (Same as subgroup 7 of group B)							
Subgroup 3			λ=10				
Steady state operation: (see 4.4.2)	1026	T_C = 150°C; I_o = 5 Adc; f = 60 Hz					
1N1614		V_{RM}(wkg) = 200 v(pk)			—	—	—
1N1615		V_{RM}(wkg) = 400 v(pk)			—	—	—
1N1616		V_{RM}(wkg) = 600 v(pk)			—	—	—
1N4458		V_{RM}(wkg) = 800 v(pk)			—	—	—
1N4459		V_{RM}(wkg) = 1,000 v(pk)			—	—	—
End points: (Same as subgroup 7 of group B)							

1/ This test shall be conducted on the initial lot and thereafter every 6 months.

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*NOTE: The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - EL
Navy - EC
Air Force - 17

Review activities:
Army - MU, MI
Air Force - 11, 19, 80
NASA - NA
DSA - ES

User activities:
Army - SM
Navy - AS, OS, MC, CG, SH

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5961-0506)

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